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Pediatric Psychiatry

Objectives:

- To learn about common pediatric psychological and psychiatric disorders
- Neurodevelopmental Disorders
- Intellectual Disability
- Autism Spectrum Disorder
- Attention-Deficit/Hyperactivity Disorder
- Conduct Disorder
- Elimination Disorders
- Enuresis
- Encopresis
- Disruptive Mood Dysregulation Disorder



Pediatric (Child) Psychiatry

- A branch of Psychiatry.
- The central focuses of the subject are behavioral and emotional disorders of childhood, but many would include physical symptoms such as non-organic headache and stomach pain in which stress or other environmental factors appear to play an important causative role.
- Delays and deviations in development, as well as general and specific learning problems lie within the practice of child psychiatry.
- Childhood period extends averagely up to age of 18 years.



Why Pediatric Psychiatry is different?

- Initiation of the consultation with the clinician.
- The stage of the development of the patient.
- Psychological problems in a child may be a manifestation of disturbance in other members of the family.
- Evidence of disturbance is based more on observation of behavior made by parents, teachers and others.
- Treatment of children makes less use of medication and other methods of individual therapies.



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Autism Spectrum Disorder

ASD is characterized by:

- 1) Persistent deficits in social communication and social interaction across multiple contexts: deficits in social-emotional reciprocity, deficits in verbal and nonverbal communicative behaviors & deficits in developing, maintaining, and understanding relationships.
2. Restricted, repetitive patterns of behavior, interests, or activities: stereotyped or repetitive motor movements, insistence on sameness, highly restricted, fixated interests & hyper or hypo-reactivity to sensory input.



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Centers for Disease Control and Prevention (CDC)

Mental health in childhood and adolescence is defined as the achievement of expected developmental, cognitive, social and emotional milestones, and by secure attachments, satisfying social relationships and effective social skills.



Data & Statistics

- **Steady Increase in Autism Prevalence (existing cases):**
- **2000 birth year of 1992- rate 1 in 150**
- **2004 birth year of 1996-rate of 1 in 125**
- **2008 birth year of 2000-rate of 1 in 88**
- **2010 birth year of 2002-rate of *1 in 68***
- **2022 birth year of 2016-rate of 1 in 44**
- **2024 birth year of 2019-rate of 1 in 36**

- **Autism Worldwide**
According to the World Health Organization (WHO), approximately 1 in 100 children worldwide has ASD.



Clinical Features

1. Impairment in social interaction: lacking social smile, fail to show the usual relatedness to their parents and other people, abnormal eye contact, ...
2. Disturbance of communication & language
3. Stereotyped behavior :the activities & play are rigid ,repetitive & monotonous. Ritualistic and compulsive phenomena are common
4. Unstability of mood
5. Abnormal response to sensory stimuli(either exaggerated or decreased)
6. Other behavioral symptoms hyperkinesis or hypokinesis ,aggressive behavior ,temper tantrums, self injurious behavior



Management & Prognosis

Prognosis is generally unfavorable

The patient needs a complicated care which include:

- 1. Educational therapy**
- 2. Behavioral therapy**
- 3. Pharmacotherapy: no specific therapy is available. It can be only symptomatic like anti-obsessive, antipsychotic and antiepileptic**



Aspects of Management

1. Management of abnormal behavior
2. Education and social services
3. Speech and language therapy ,occupational therapy, dietary advice...
4. Treat medical conditions (e.g. epilepsy ,GIT problems)
5. Help for families
6. Pharmacotherapy :symptom management (e.g. antipsychotic for stereotypes),SSRI for compulsive and self harming behaviors and depression, and anxiety



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Aspergers Syndrome(AS)

A syndrome first described by Hans Asperger's in 1944 ,and sometimes called autistic psychopath ,is characterized by:-

- The child develops normally until about the third year when they begin to lack warmth in relationships, and speak in monotonous stilted ways
- Severe persistent impairment in social interactions ,repetitive behavior patterns, and restricted interests
- IQ and language are normal or in some cases, superior
- Motor mannerisms such as hand and finger twisting, or whole body movements



- They are often clumsy and eccentric
- They are more interested in others than autistic children
- The disorder is more common in boys than girls
- A family history of autism may be present
- The cause of AS is unknown
- It differ from autism in that there is no general delay or retardation of cognitive development or language
- They are solitary, and embark on and spend much time in narrow interests



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Attention-Deficit Hyperactivity Disorder

This disorder is common ,appears more often in boys than in girls and causes disruption in school and at home

It is characterized by:

- 1. Features of hyperactivity: age-inappropriate hyperactivity which is mostly purposeless & intolerable causing a lot of disturbance**
- 2. Poor attention span**
- 3. Impulsivity**

These symptoms should be present for at least 6 months before the diagnosis is made

The symptoms should be present in more than one setting (home, school, work)

And should be severe enough to cause significant impairment



ADHD

- 3%-7% of children suffer from ADHD
- ADHD is diagnosed approximately three times more often in boys than in girls
- As many as half of those with ADHD also have other mental disorders
- Over half of the children diagnosed with ADHD carry the disorder into adulthood
- A large number of adults who were never diagnosed as a child show clear symptoms of ADHD



ADHD

- The symptoms of ADHD are present since the early childhood (before the age of 7 years)
- The causes of ADHD are unknown ,but the disorder is predictably associated with a variety of other disorders that affect the brain function ,such as learning disorders
- The suggested contributory factors to ADHD include prenatal toxic exposure, prematurity, and prenatal mechanical insult to the fetal nervous system
- Food additives ,colorings, preservatives, and sugar have been suggested as possible causes
- There is evidence for a genetic cause



Inattention symptoms 6+

1. Often fails to give close attention to details or makes careless mistakes in schoolwork, work, or other activities
2. Often has difficulty sustaining attention in tasks or play activities
3. Often does not seem to listen when spoken to directly
4. Often does not follow through on instructions and fails to finish school-work, chores, or duties in the workplace (not oppositional)



Inattention symptoms 6+

5. Often has difficulty organizing tasks and activities
6. Often avoids, dislikes, or is reluctant to engage in tasks that require a sustained mental effort
7. Often loses things necessary for tasks or activities
8. Often easily distracted by extraneous stimuli
9. Often forgetful in daily activities



Hyperactivity

1. Often fidgets with hands or feet or squirms in seat
2. Often leaves seat in classroom or in other situations in which remaining seated is expected
3. Often runs about or climbs excessively in situations in which it is inappropriate (adolescents ~ may be subjective feelings of restlessness)
4. Often has difficulty playing or engaging in leisure activities quietly
5. Often “on the go” or often acts as if “driven by a motor”
6. Often talks excessively



Impulsivity

1. Often blurts out answers before questions have been completed
2. Often has difficulty awaiting turn
3. Often interrupts or intrudes on others



Treatment of ADHD

1. **Pharmacotherapy:**

- a. CNS stimulants: dextroamphetamine, methylphenidate, and pemoline
- b. Antidepressants
- c. Atomoxetine: new non-stimulant anti ADHD

2. **Psychotherapy :**

which include behavioral therapy , education of parents and teachers



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Conduct Disorder

- Disorder of behaviour characterised by repetitive and persistent pattern of dissocial, aggressive or defiant conduct.
- Not just delinquency
- Problem behaviours vary with development



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Conduct Disorder

Under the age of 5 years

- aggression - physical and verbal
- destructiveness
- poor attention and concentration
- frequent, severe tantrums



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Conduct Disorder



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Conduct Disorder

Conduct at ages 5-12 years:

- lying
- stealing
- defiance
- disruption in school
- cruelty to animals
- fire setting
- solvent abuse



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Conduct Disorder

Conduct in Adolescents

- truancy
- delinquency
- violence
- sex offences
- drug/alcohol/substance abuse



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Conduct Risk Factors

- boys >> girls
- inner cities > rural areas
- socio-economic disadvantage
- family conflict and poor communication in the family
- “difficult” temperament
- Specific reading disability



Treatment of Conduct

- There is no specific pharmacological intervention for this disorder.
- What can be offered, are limited to special training programs for both the child and parents, anger management, remedial teaching or residential care for severe uncontrolled cases.



Elimination Disorders

I. Enuresis :

The repeated voiding of urine into clothes or bed ,whether, the voiding is involuntary or intentional .The behavior must occur twice weekly for at least 3 months or must cause clinically significant distress or impairment socially or academically.The child's age must be at least 5 years



Treatment of Enuresis

1. Reassurance of the child.
2. Restriction of fluid intakes before bedtime.
3. Regular wakeup of the child to urinate during night times (e.g., at midnight, and 03:00 am).
4. Enuresis alarm.
5. Medications: antidiuretic hormones like (Desmopressin) in forms of tablets or nasal sprays, and/or Imipramine (TCAD)



2. **Encopresis:**

Passing feces into inappropriate places whether the passage is involuntary or intentional. The pattern must be present for at least 3 months ; the child's age must be at least 4 years

Treatment: There is no specific pharmacological intervention. Child reassurance, toilet training, and behavioral modifications may be satisfactory.



Depressive Disorder

Depressive disorders

- Any kind of mood disorder may occur in children, yet with possible different clinical manifestations.
- However, DSM-5 introduced a new depressive disorder, „Disruptive Mood Dysregulation Disorder“, with onset between 6 to 18 years old.



DISRUPTIVE MOOD DYSREGULATION DISORDER

- Severe recurrent temper outbursts manifested verbally and/or behaviorally

A. Verbal rages

B. Physical aggression toward people or property

- Temper outbursts are inconsistent with developmental level
- Temper outbursts occur 3+ times per week



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DISRUPTIVE MOOD DYSREGULATION DISORDER

Mood between temper outbursts is:

- Persistently irritable or angry most of the day
- Nearly every day
- Observable by others



Treatment

- Since this diagnosis is newly introduced in the DSM, clear information about possible etiological factors, courses and development, and treatments are lacking yet.
- Nonetheless, DSM referred to the possibilities of progression of children with these symptoms pattern to adulthood anxiety or depressive disorder.



Intellectual Disability

- Over the years, several terms have been applied to people with intellectual impairment from early life.
- In the nineteenth and early twentieth centuries, the word '**idiot**' was used for people with severe intellectual impairment, and '**imbecile**' for those with moderate impairment.
- The special study and care of such people was known as **the field of mental deficiency**.
- When these words came to carry stigma, they were replaced by the terms **mental subnormality** and **mental retardation**.
- The term **mental handicap** has also been widely used, but now the term **learning disability** is generally preferred in the UK.



- Historically, ID has been defined relying on Intelligence Quotient (IQ) scores of below 70.
- Likewise, the severity of ID was determined according to the same measure.
- However, although the name, ID, denotes a disabled intelligence, the disability is not limited to the intellectual abilities alone; further, there are adaptive functioning deficits in conceptual, social, and practical domains.



DSM-5 criteria for diagnosis

- A. Deficits in intellectual functions, such as reasoning, problem solving, planning, abstract thinking, judgment, academic learning, and learning from experience, confirmed by both clinical assessment and individualized, standardized intelligence testing.
- A. Deficits in adaptive functioning that result in failure to meet developmental and sociocultural standards for personal independence and social responsibility.
- B. Onset of intellectual and adaptive deficits during the developmental period.



Types of ID

A. According to severity: the severity of ID was determined in the past solely by the scores of IQ as follows:

1. Mild: $IQ = 55-70$.
2. Moderate: $IQ = 40-55$.
3. Severe: $IQ = 25-40$.
4. Profound: $IQ < 25$.



B. According to underlying genetic syndromes:

1. Syndromal ID: ID might be a feature of prenatally determined genetic syndromes, like Down (trisomy 21) syndrome, Lesch-Nyhan syndrome, Rett's syndrome, etc...
2. Non-syndromal ID: here, the neurodevelopmental disorder is acquired due to prenatal infections or toxins, perinatal events-related encephalopathies, or early postnatal infections, exposure to toxins, trauma, hypoxia, or infantile seizure disorders.



Associated symptoms

- ID frequently coupled with other behavioral and psychological symptoms like psychosis, depression, aggression, or disinhibitions.
- Furthermore, there may be co-morbid psychiatric disorders like schizophrenia, for instance, which is more prevalent among ID population compared to general population



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Epidemiology

General population prevalence is about 1%. Vast majority of IDs are of „Mild“ severity. The prevalence is thought to be more among male population.



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Intervention

Special Education Services for Child/Adolescent

Comprehensive program that addresses:

- Adaptive Skills Training
- Social Skills Training
- Vocational Training
- Group Therapy ~ Practice managing hypothetical real-life problems while receiving supportive feedback



Behavioral and Cognitive Therapy

- Positive reinforcement for desired behaviors and punishment for objectionable behaviors
- Relaxation exercises with self-instruction



Family Education

- Education about methods to enhance child's competence and self-esteem while maintaining realistic expectations
- Education regarding balance between fostering independence and providing a supportive environment
- Encourage family members participation in psychotherapy
Express guilt, despair, anger, frustration
- Provide family members with basic and current medical information regarding causes and treatment of mental retardation



Pharmacological Intervention

- Aggression and Self-injurious Behaviors
- Stereotypical Motor Movements
- Explosive Rage Behavior
- Attention-Deficit/Hyperactivity Disorder
- Emotional Concerns (e.g., Depression, Anxiety, Etc.)



Learning Disorders

Diagnostic Criteria

- A. Difficulties learning and using academic skills, as indicated by the presence of at least one of the following symptoms that have persisted for at least 6 months, despite the provision of interventions that target those difficulties:
1. Inaccurate or slow and effortful word reading (e.g., reads single words aloud incorrectly or slowly and hesitantly, frequently guesses words, has difficulty sounding out words).
 2. Difficulty understanding the meaning of what is read (e.g., may read text accurately but not understand the sequence, relationships, inferences, or deeper meanings of what is read).
 3. Difficulties with spelling (e.g., may add, omit, or substitute vowels or consonants).
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4. Difficulties with written expression (e.g., makes multiple grammatical or punctuation errors within sentences; employs poor paragraph organization; written expression of ideas lacks clarity).
 5. Difficulties mastering number sense, number facts, or calculation (e.g., has poor understanding of numbers, their magnitude, and relationships; counts on fingers to add single-digit numbers instead of recalling the math fact as peers do; gets lost in the midst of arithmetic computation and may switch procedures).



6. Difficulties with mathematical reasoning (e.g., has severe difficulty applying mathematical concepts, facts, or procedures to solve quantitative problems).
- B. The affected academic skills are substantially and quantifiably below those expected for the individual's chronological age, and cause significant interference with academic or occupational performance, or with activities of daily living, as confirmed by individually administered standardized achievement measures and comprehensive clinical assessment. For individuals age 17 years and older, a documented history of impairing learning difficulties may be substituted for the standardized assessment.
- C. The learning difficulties begin during school-age years but may not become fully manifest until the demands for those affected academic skills exceed the individual's limited capacities (e.g., as in timed tests, reading or writing lengthy complex reports for a tight deadline, excessively heavy academic loads).
- D. The learning difficulties are not better accounted for by intellectual disabilities, uncorrected visual or auditory acuity, other mental or neurological disorders, psychosocial adversity, lack of proficiency in the language of academic instruction, or inadequate educational instruction.



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Thank You

